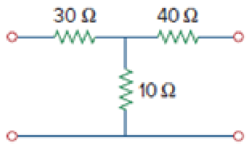
Chapter 19, Problem 67P

Problem

If three copies of the circuit in Fig. 19.114 are connected in parallel, find the overall transmission parameters.

Figure 19.114



Step-by-step solution

Step 1 of 6

Consider the following circuit:

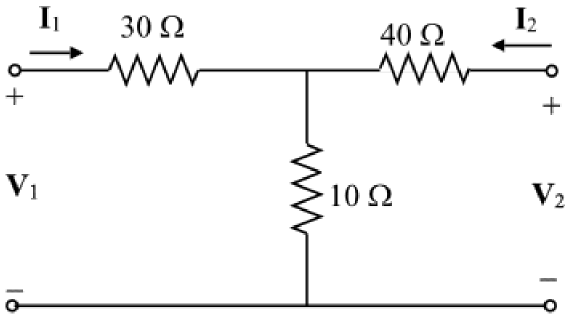


Figure 1

Step 2 of 6

Define *z* parameters as follows.

$$\mathbf{V}_1 = \mathbf{z}_{11}\mathbf{I}_1 + \mathbf{z}_{12}\mathbf{I}_2$$
$$\mathbf{V}_2 = \mathbf{z}_{21}\mathbf{I}_1 + \mathbf{z}_{22}\mathbf{I}_2$$

$$\mathbf{z}_{11} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{12} = \left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}$$

$$\mathbf{z}_{21} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0}$$

$$\mathbf{z}_{22} = \left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0}$$

To determine $\mathbf{z}_{11}$ and $\mathbf{z}_{21}$, apply voltage source $\mathbf{V}_1$ and leave the output port open.

Apply Kirchhoff's voltage law to the input loop.

$$\mathbf{V}_1 = (30+10)\mathbf{I}_1$$
$$\mathbf{V}_1 = 40\mathbf{I}_1$$
$$\left.\frac{\mathbf{V}_1}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0} = 40$$

Apply Kirchhoff's current law to the top node.

$$\frac{\mathbf{V}_1 - \mathbf{V}_2}{30} = \frac{\mathbf{V}_2}{10}$$
$$\mathbf{V}_1 - \mathbf{V}_2 = 3\mathbf{V}_2$$
$$\mathbf{V}_1 - 4\mathbf{V}_2 = 0$$

Substitute $40\mathbf{I}_1$ for $\mathbf{V}_1$.

$$40\mathbf{I}_1 - 4\mathbf{V}_2 = 0$$
$$40\mathbf{I}_1 = 4\mathbf{V}_2$$
$$\left.\frac{\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{I}_2=0} = 10$$

Step 3 of 6

To determine $\mathbf{z}_{12}$ and $\mathbf{z}_{22}$, apply voltage source $\mathbf{V}_2$ to the output port and leave the input port open.

Apply Kirchhoff's voltage law to the output loop.

$$\mathbf{V}_2 = (40+10)\mathbf{I}_2$$
$$\mathbf{V}_2 = 50\mathbf{I}_2$$
$$\left.\frac{\mathbf{V}_2}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} = 50$$

And

$$\mathbf{V}_1 = 10\mathbf{I}_2$$
$$\left.\frac{\mathbf{V}_1}{\mathbf{I}_2}\right|_{\mathbf{I}_1=0} = 10$$

The z parameters of the network are $\begin{bmatrix} 40 & 10 \\ 10 & 50 \end{bmatrix}$.

Step 4 of 6

For parallel connection of three networks as in Figure 1, y-parameters are equal to 3 times that for a single circuit.

Determine y-parameters.

Write the conversion formulae to find y-parameters.

$$y_{11} = \frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}}$$

$$y_{12} = -\frac{z_{12}}{z_{11}z_{22} - z_{12}z_{21}}$$

$$y_{21} = -\frac{z_{21}}{z_{11}z_{22} - z_{12}z_{21}}$$

$$y_{22} = \frac{z_{11}}{z_{11}z_{22} - z_{12}z_{21}}$$

Substitute z parameter values.

$$\begin{aligned} y_{11} &= \frac{z_{22}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= \frac{50}{(40)(50)-(10)(10)} \\ &= \frac{50}{1900} \\ &= \frac{5}{190} \end{aligned}$$

$$\begin{aligned} y_{12} &= -\frac{z_{12}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= -\frac{10}{(40)(50)-(10)(10)} \\ &= -\frac{10}{1900} \\ &= -\frac{1}{190} \end{aligned}$$

$$\begin{aligned} y_{21} &= -\frac{z_{21}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= -\frac{10}{(40)(50)-(10)(10)} \\ &= -\frac{10}{1900} \\ &= -\frac{1}{190} \end{aligned}$$

$$\begin{aligned} y_{22} &= \frac{z_{11}}{z_{11}z_{22} - z_{12}z_{21}} \\ &= \frac{40}{(40)(50)-(10)(10)} \\ &= \frac{40}{1900} \\ &= \frac{4}{190} \end{aligned}$$

Step 5 of 6

Determine *y* parameters of the parallel network.

$$y_{11} = 3\left(\frac{5}{190}\right)$$
$$= \frac{15}{190}$$
$$y_{12} = 3\left(-\frac{1}{190}\right)$$
$$= -\frac{3}{190}$$
$$y_{21} = 3\left(-\frac{1}{190}\right)$$
$$= -\frac{3}{190}$$
$$y_{22} = 3\left(\frac{4}{190}\right)$$
$$= \frac{12}{190}$$

Step 6 of 6

Convert *y* parameters to transmission parameters using the following conversion formulae.

$$\mathbf{A} = -\frac{y_{22}}{y_{21}}$$
$$\mathbf{B} = -\frac{1}{y_{21}}$$
$$\mathbf{C} = -\frac{\Delta_y}{y_{21}}$$
$$\mathbf{D} = -\frac{y_{11}}{y_{21}}$$

Enter the following code in MATLAB to determine the *T* parameters.

```
>> Y=[15/190 -3/190;-3/190 12/190]

Y =

0.0789 -0.0158

-0.0158 0.0632

>> dy=det(Y)

dy =

0.0047

>> T=[-Y(2,2)/Y(2,1) -1/Y(2,1);-dy/Y(2,1) -Y(1,1)/Y(2,1)]

T =

4.0000 63.3333

0.3000 5.0000
```

Thus, the overall transmission parameters of the parallel network are $\boxed{\begin{bmatrix} 4 & 63.33 \\ 0.3 & 5 \end{bmatrix}}$